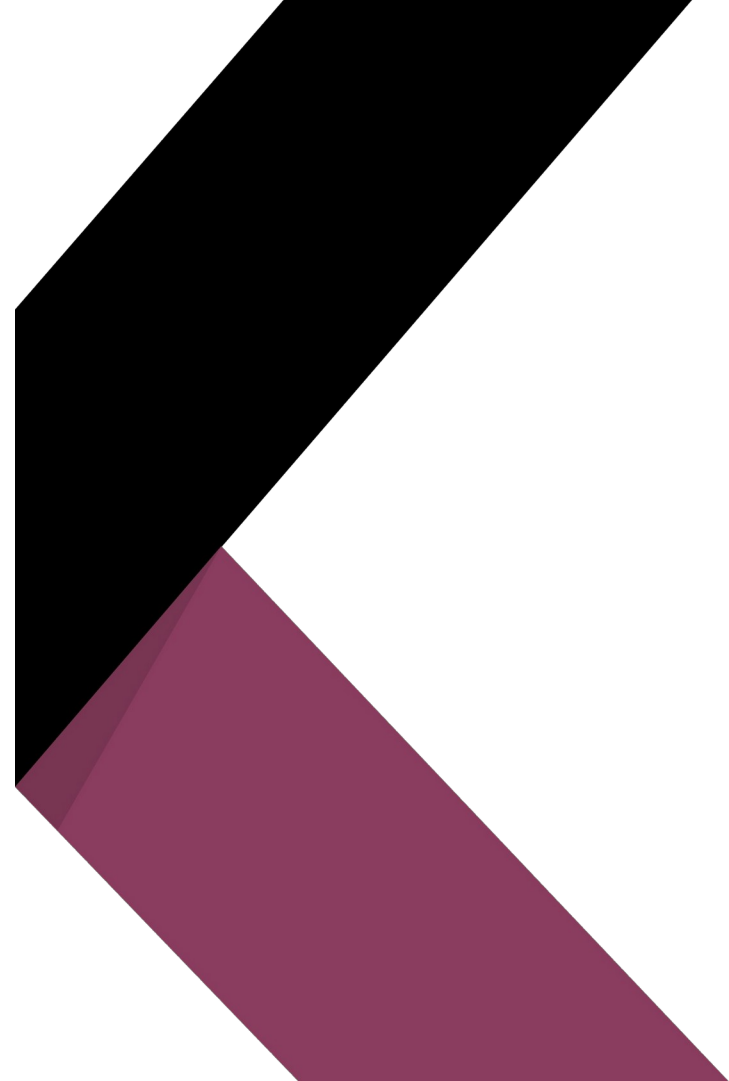


E/E Architecture

Vehicle





E/E Architecture



Background

“Today an average new automobile includes more than 40 electronic controllers, five miles of wiring, and more than 10 million lines of software code. Electronics make up nearly 40% of the content of today’s average new automobile, and their share will continue to grow.”

*Thomas H. Klier, senior economist, and
James M. Rubenstein, professor, Miami University*



**Safety and
driving assistance**



**Information,
comfort & convenience**



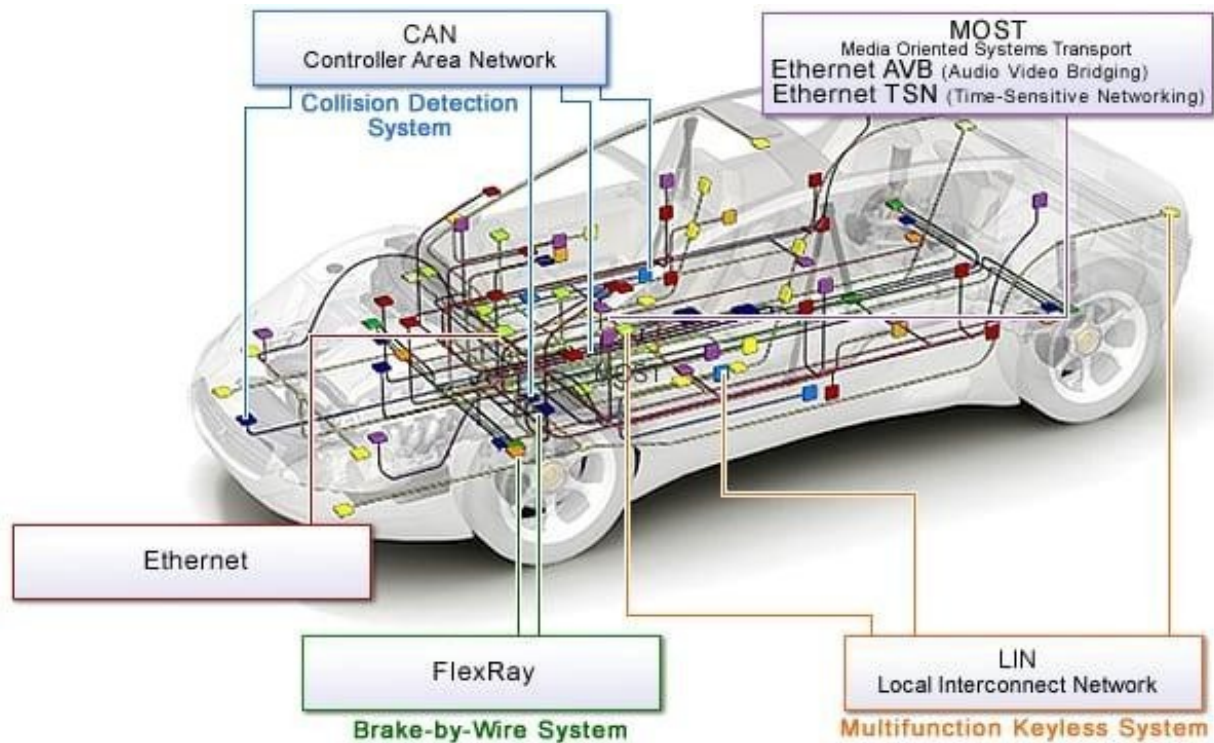
**Navigation and
connectivity**



**Fuel economy and
Electrification**



Quick Overview

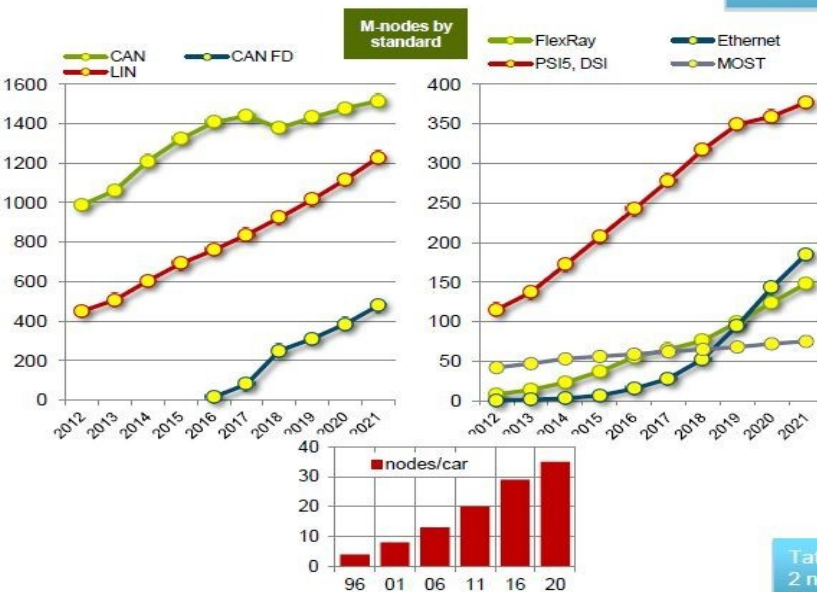
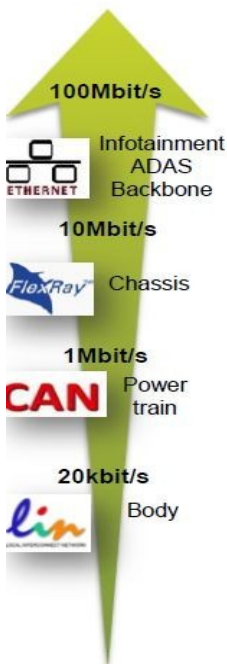




Market trends

In-Vehicle Networking

A Fast Growing, Complex, Differentiated Market



Daimler S-Class
>100 nodes



Different
OEMs,
different
complexity,
needs,
solutions



Tata Nano
2 nodes

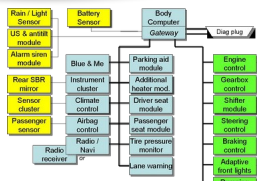




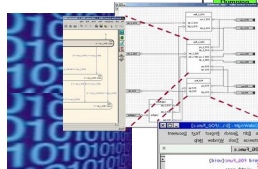
EE Architecture – is / is Not

Technologies & Standards

Network architecture



Software architecture



Diagnosis Standard

Hardware standards



Components

Body Electronic Modules



Power Distribution Units



Wiring topology & connectors



+ Interface to all EE systems:

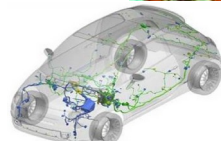


Vehicle application tasks

Analyze need to add/remove components



Packaging and wiring lay-out



New software development & test



Tuning SW to reach performance



Electrical & Electronic Architecture (EEA) is the heart of functional integration of the vehicle, distributing power/signals/functions to all EE systems



E/E Architecture Definition & Characteristics

E/E Architecture is usually defined by means of:

- ECUs/Network Topology
- Network & Diagnostic protocol
- CAN Standard map
- Electrical interfaces (sensors, pedal assembly, starter)
- ECU re-programming & Vehicle Configuration Management
- Cyber Security level / AUTOSAR compliancy

EE Architecture Characteristic	Purpose
EEA Standards	Engineering Standards - - EMC, Electrical Systems, Networks, Diagnostics, Vehicle Configuration Management, Development Processes, ...
Network Topology	Defines overall communications network within the vehicle including which ECUs are on which communication bus and the type of power feed used for those ECUs..
SLA (Sense / Logic / Actuation)	Defines distribution of feature and function including inputs and outputs
Electrical Interface	Defines hardware electrical interfaces between ECUs and switches, sensors, actuators and other ECUs
Network Management (related to AUTOSAR)	Defines networking "wake-up" and "sleep" management
Diagnostic Requirements	Defines basic diagnostic specs, flash bootloader and diagnostic services and protocol
Vehicle Configuration Management	Defines the methodology used to communicate vehicle specific content (standard and optional features) to the vehicle ECUs thereby reducing module part number complexity
Network Database	Defines CAN and LIN message IDs, data length, timing, and signal content and location
VF (Vehicle Functions)	Detailed description of the functionality, exchanged signals, and algorithms related to a particular feature or sub-feature



CAN Sleep/Wake up

The CAN-C1, CAN-C2 and CAN-BH networks enter Sleep mode (fall asleep) after about 10 ÷ 12 seconds from the moment the key has been turned OFF.



The networks "wake up" with the key in the OFF position when one of the car doors changes from the "Closed" to the "Open" state.



Key Drivers for E/E Architecture Development

The following items are key drivers in a Network Architecture:

1. Integration Functions Complexity
 - Complexity/magnitude of new requirements are escalating at an exponential rate
2. Bus Load Capacity
 - Current architecture networks are running at 50-60% bus load (C-CAN)
 - To ensure robust communication (no missed message or latency issues), networks are maintained below a threshold of 65-70% bus load
 - Extrapolating the expected bus loads based on new features with new bus message requirements, the current C-CAN bus could exceed bus load limits if not properly managed
3. Cyber Security
 - Carmakers must protect its vehicles against Cyber attacks. Secure CAN communication requires message authentication (message data increases by 50% to support message authentication)
 - Safety critical modules will require hardware Trust Anchors (as many as 18 ECUs on high feature content vehicles)
4. AUTOSAR
 - AUTOSAR is an open and standardized automotive SW architecture which allows SW exchangeability between vehicle platform, supplier's solutions, OEM's applications
 - Main benefits: Formal & Defined Architecture, Hardware Independence, Software Interfacing Standardization, Functional Safety support





VF (Vehicle Functions)

▪ Vehicle Functions

- Functionalities that need an interaction (usually via CAN/LIN networks) of different Electronic Control Units (ECUs) inside a vehicle
- VFs include high level integration requirements among vehicle ECUs
- VFs are specific for each car line and for each E/E architecture

1.8.3 B/BH-CAN

STATUS_BH BCM2.BatteryChargeSts
Battery State of Charge (in %), as acquired from the battery sensor.

1.8.4 C-CAN

1.8.5 LIN

IBS_Slow2.IBS3_Q_Received
IBS_Slow2.IBS3_Q_Released
IBS_Slow2.IBS3_TmLstResetDays
IBS_Slow2.IBS3_TmLstResetSec
IBS_Slow2.IBS3_BatteryId_chosen
IBS_Wakeup.IBS3_Flag_Disconnect
IBS_Wakeup.IBS3_HW_version
IBS_Wakeup.IBS3_Ibatt_Keyoff_Dsch
IBS_Wakeup.IBS3_Initialized
IBS_Wakeup.IBS3_LowBatteryFlag

1 IBS 3 Management [VF512_V4_R2]

1.1 Vehicle Function Data

1.1.1 Vehicle Function Area

Energy

1.1.2 Vehicle Function Group

Energy

1.1.3 Vehicle Function Owner

Marco Sergi

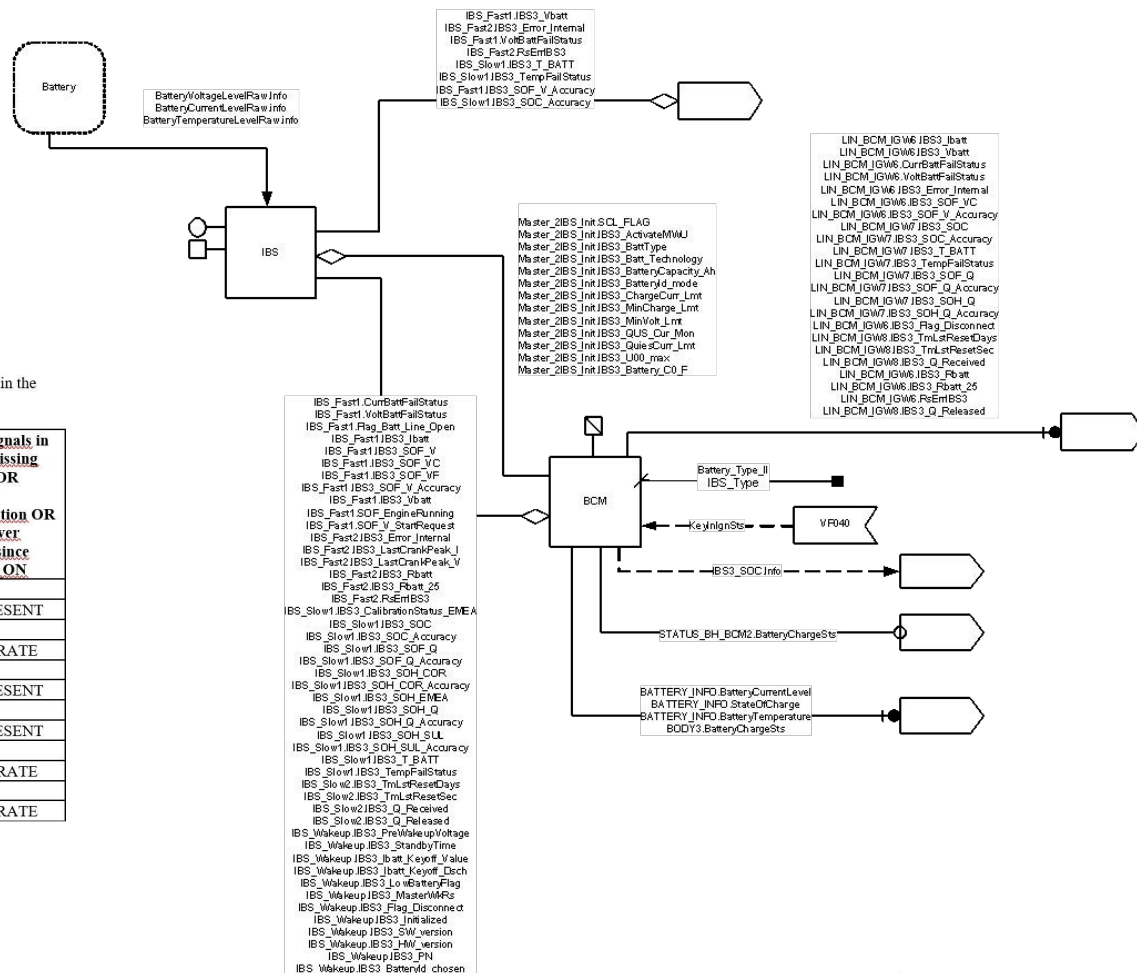
1.2 Revision Notes

Date	Author	Description
2018/10/19	Marco Sergi	First release
2020/06/12	Marco Sergi	CR14710: - Par 1.10.1 Ignition Working Conditions Table: Modify condition of Ignition OFF - Par. 1.11.1.3: Wake Up functions: add note about wake up in Key off condition.

This document is based on template "VF Template 9.2".



VF (Vehicle Functions)



1.11.2.3 Gateway Function

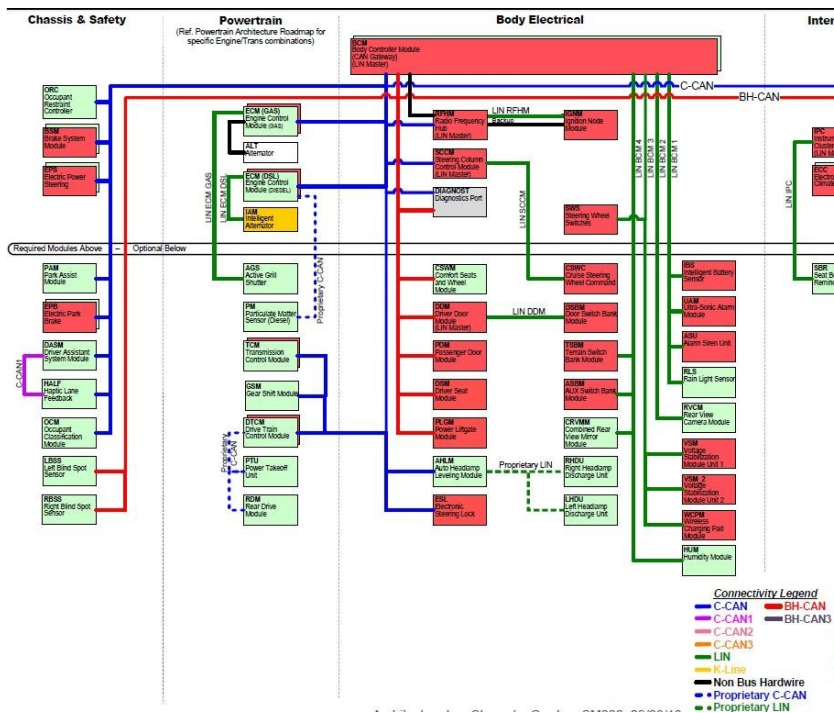
Part of the signals coming from the IBS via LIN are broadcasted in the B-CAN and/or C-CAN, as specified in the following tables:

LIN Signal	C-CAN Signals	C Can Signals in case of Missing message OR LOSS communication OR Signal never received since first Key_ON
IBS_Fast1_IBS3_Ibatt	LIN_BCM_IGW6_IBS3_Ibatt	SNA
IBS_Fast1_CurrBattFailStatus	LIN_BCM_IGW6_CurrBattFailStatus	FAIL PRESENT
IBS_Slow1_IBS3_SOC	LIN_BCM_IGW7_IBS3_SOC	SNA
IBS_Slow1_IBS3_SOC_Accuracy	LIN_BCM_IGW7_IBS3_SOC_Accuracy	INACCURATE
IBS_Fast1_IBS3_Vbatt	LIN_BCM_IGW6_IBS3_Vbatt	SNA
IBS_Fast1_VoltBattFailStatus	LIN_BCM_IGW6_VoltBattFailStatus	FAIL PRESENT
IBS_Slow1_IBS3_T_BATT	LIN_BCM_IGW7_IBS3_T_BATT	SNA
IBS_Slow1_IBS3_TempFailStatus	LIN_BCM_IGW7_IBS3_TempFailStatus	FAIL PRESENT
IBS_Slow1_IBS3_SOF_Q	LIN_BCM_IGW7_IBS3_SOF_Q	SNA
IBS_Slow1_IBS3_SOF_Q_Accuracy	LIN_BCM_IGW7_IBS3_SOF_Q_Accuracy	INACCURATE
IBS_Fast1_IBS3_SOF_VC	LIN_BCM_IGW6_IBS3_SOF_VC	SNA
IBS_Fast1_IBS3_SOF_V_Accuracy	LIN_BCM_IGW6_IBS3_SOF_V_Accuracy	INACCURATE

CUSW

0 Key Features:

- **Star** topology, BCM central hub
- **11bit** CAN IDs
- LIN **2.x** protocol
- **Direct** PROXI in ECM
- IPC stays on both B-CAN+**C-CAN**
- Cruise commands **LIN** connected to BCM
- **Signal** key architecture
- **Advanced** frame security mechanism (**CRC/MC** coverage)
- Torque CAN interface at **flywheel** torque level (in Nm)



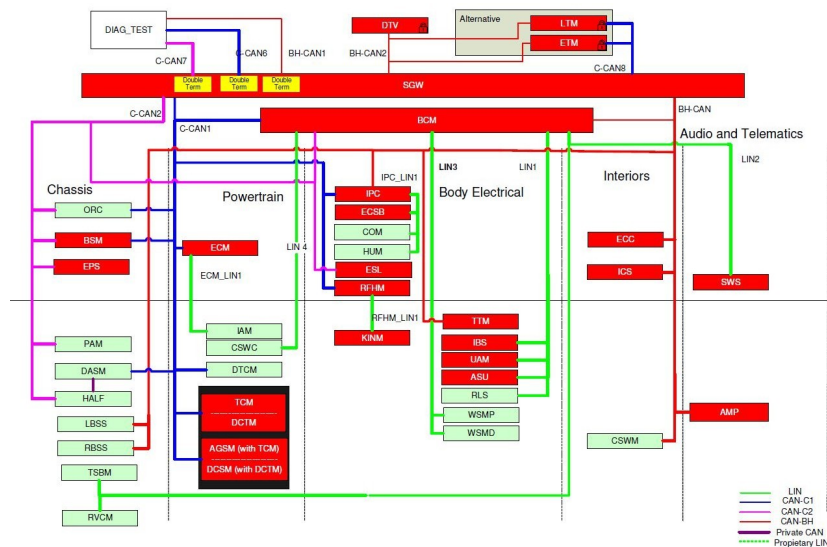


Network Topology Atlantis/Nextgen

○ Example: 520

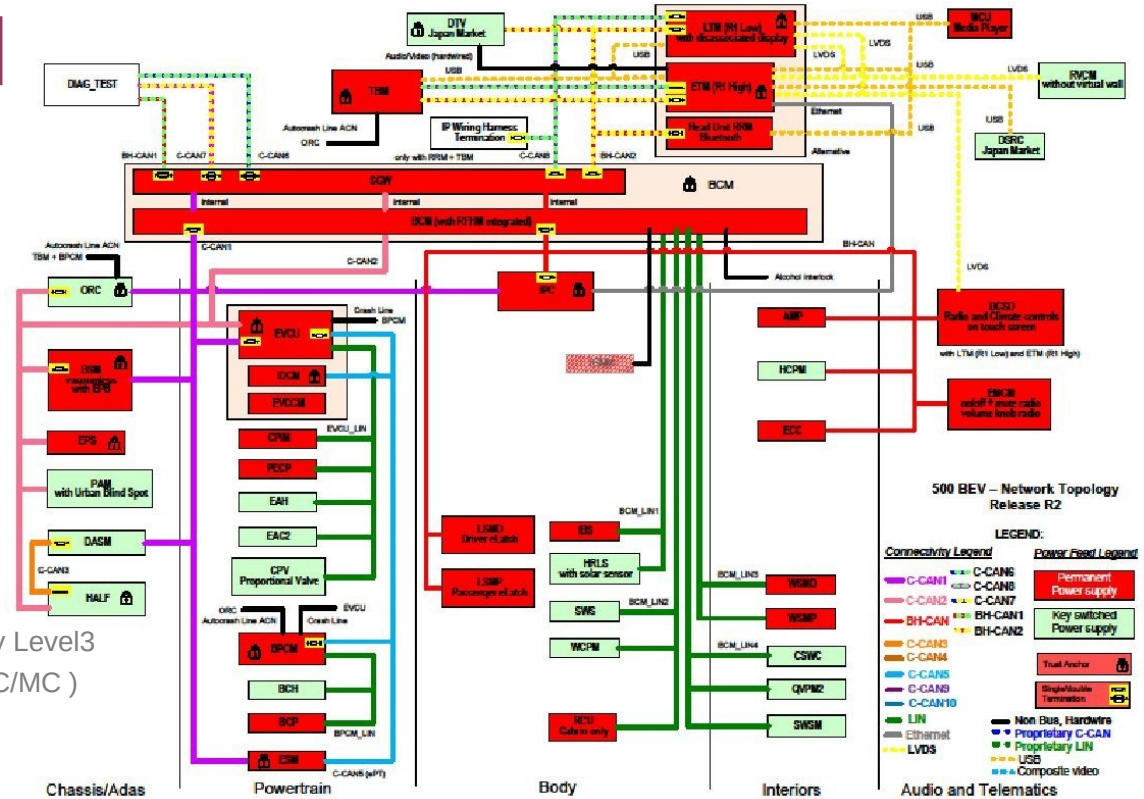
○ Key Features:

- Daisy chain topology, ECM/BCM terminator
- 11bit CAN IDs
- LIN 2.x protocol
- Direct PROXI in ECM
- IPC stays on both B-CAN+C-CAN
- Cruise commands LIN connected to BCM
- Signal key architecture
- Advanced frame security mechanism (CRC/MC coverage)
- Torque CAN interface at flywheel torque level (in Nm)
- **Multiple C-CANs** + New Tx types for busload reduction
- **AUTOSAR** compliancy (only J4A - Atlantis Tier1)
- Secure Gateway (**SGW**) for Cyber Security Level3



BEV

- **Star** topology, BCM central hub
- 11bit CAN IDs
- LIN 2.x protocol
- IPC stays on C-CAN
- Cruise commands LIN connected to BCM
- Signal key architecture
- Multiple C-CANs + **Multiple** LINs
- Dedicated C-CAN for e-PT (**HEV**)
- Secure Gateway (SGW) for Cyber Security Level3
- Advanced frame security mechanism (CRC/MC)





Vehicle network CAN

The main CAN bus that are used on the vehicle are classified on the basis of the baud rate.

The most used network are:

- C-CAN: the baud rate is 500 kb/s;
- B-CAN: the baud rate is 50 kb/s;
- BH-CAN: the baud rate is 125 kb/s;
- FD-CAN: the baud rate is variable up to 15Mb/s.

Usually on the C-CAN there are for sure all the ECUs that are emission relevant in accord to EOBD normative.

Usually on the BH-CAN there are all the ECUs that manage multimedial information.



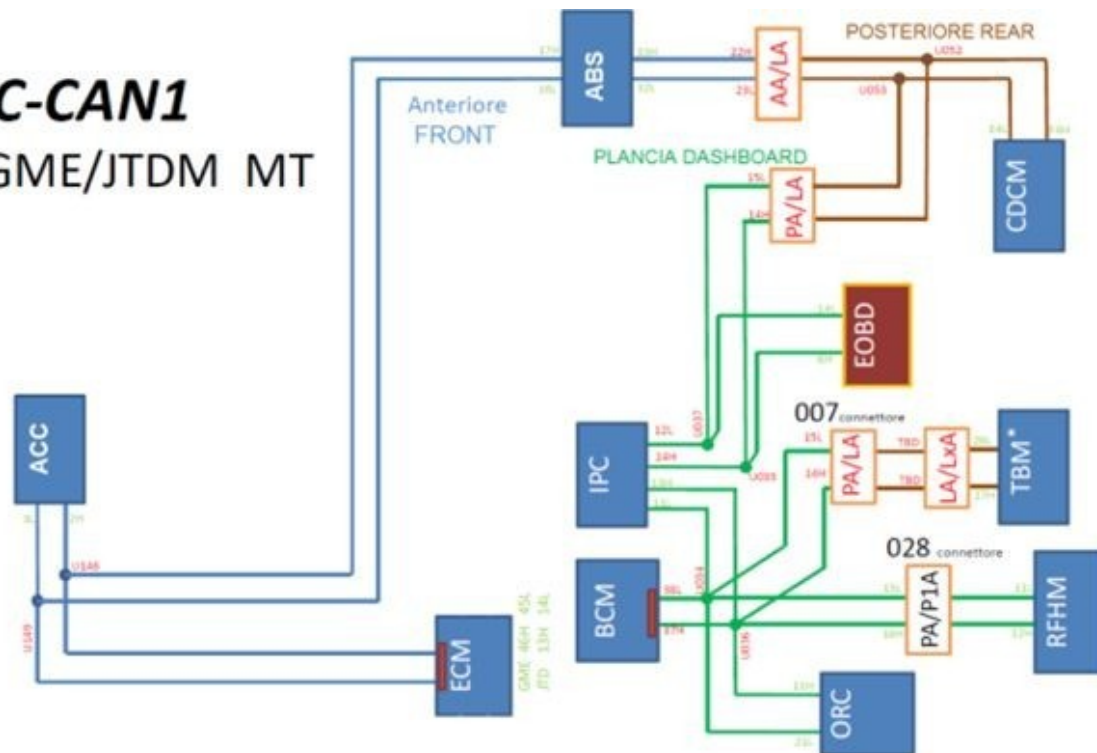
Vehicle network **Nextgen**

The car is equipped with an electrical / electronic architecture called Next-Generation. Data transmission between the various ECUs in the Next-Generation architecture takes place via the following three digital networks:

- ☐ CAN-C1 (high speed 500 Kb / s)
- ☐ CAN-C2 (high speed 500 Kb / s)
- ☐ CAN-BH (medium speed 125 Kb / s)

C-CAN1
GME/JTDM MT

EOBD (Presa Di Diagnosi)





Can-C1 GME AT

BCM (Body Control Module)

IPC (Instrument Panel Cluster – Quadro strumenti)

ORC (Occupant Restraint Control – Modulo airbag)

RFHM (Radio Frequency Module – Modulo Radio frequenza)

ABS (Antilock Breaking System Module – Modulo ABS)

ACC (Adattative Cruise Control – Modulo Cruise control adattativo)

TBM* (Telematic Box Module – Modulo Telematico)

ECM (Engine Control Module - Modulo controllo motore)

CDCM (Chassis Domain Control Module Modulo controllo attivo della dinamica)

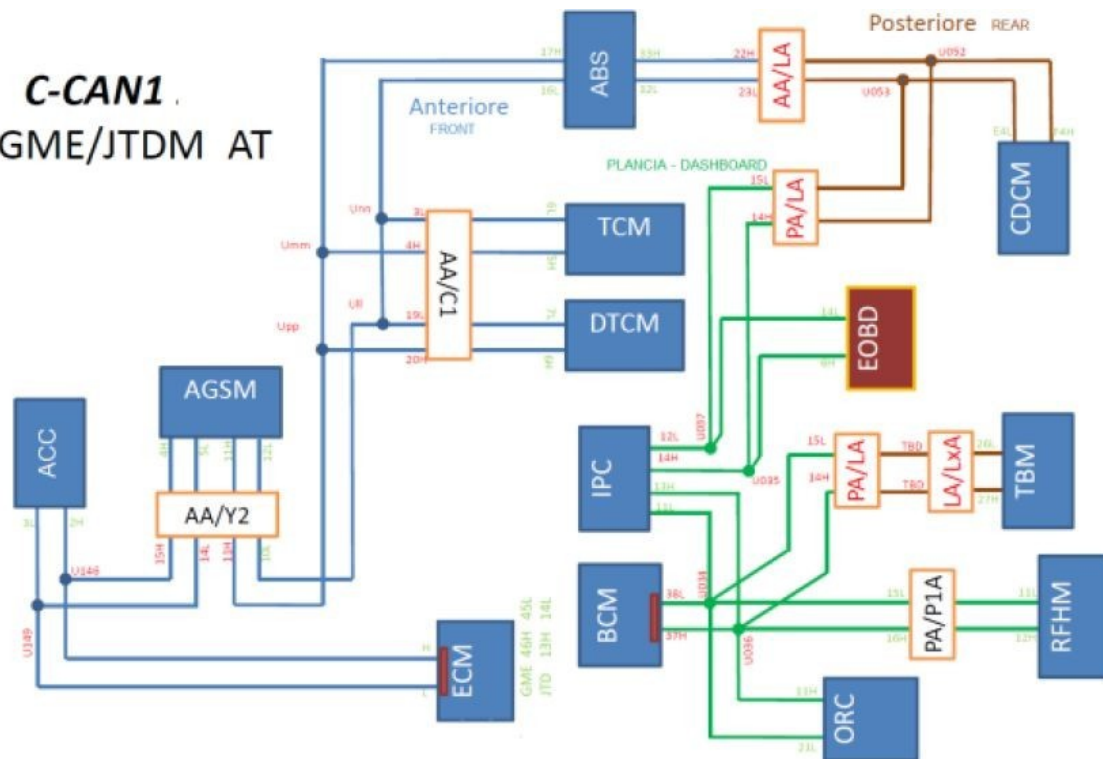
TCM (Transmission Control Module – Modulo cambio automatico)

DTCM (Drive Train Control Module – modulo controllo trazione AWD)

AGSM (Automatic Gearbox Shifter Module – Modulo leva selezione cambio automatico)

EOBD (Presa Di Diagnosi)

C-CAN1 GME/JTDM AT





CAN-C2.

The CAN-C2 network is the data exchange location of the electronic modules (ECU) in the chassis area:

ABS (Antilock Breaking System – Modulo ABS)

BCM (Body Control Module)

ESL (Electric Steering Lock – Modulo Bloccasterzo)

PAM (Parking Aid Module – Module assistenza al parcheggio)

EPS (Electric Power Steering – Modulo Guida Elettrica.)

HALF (Haptic Lane Feedback – Modulo di assistenza al mantenimento di corsia)

ORC (Occupant Restraint Control - Airbag)

TBM* (Telematic Box Module – Modulo Telematico)

AAML (Active Aerodynamic Module Left – Attuatore sinistro di movimentazione Splitter anteriore)

AAMR (Active Aerodynamic Module Right – Attuatore destro di movimentazione Splitter anteriore)

AFLS (Adaptive Front Light System – Modulo fari autoadattativi)

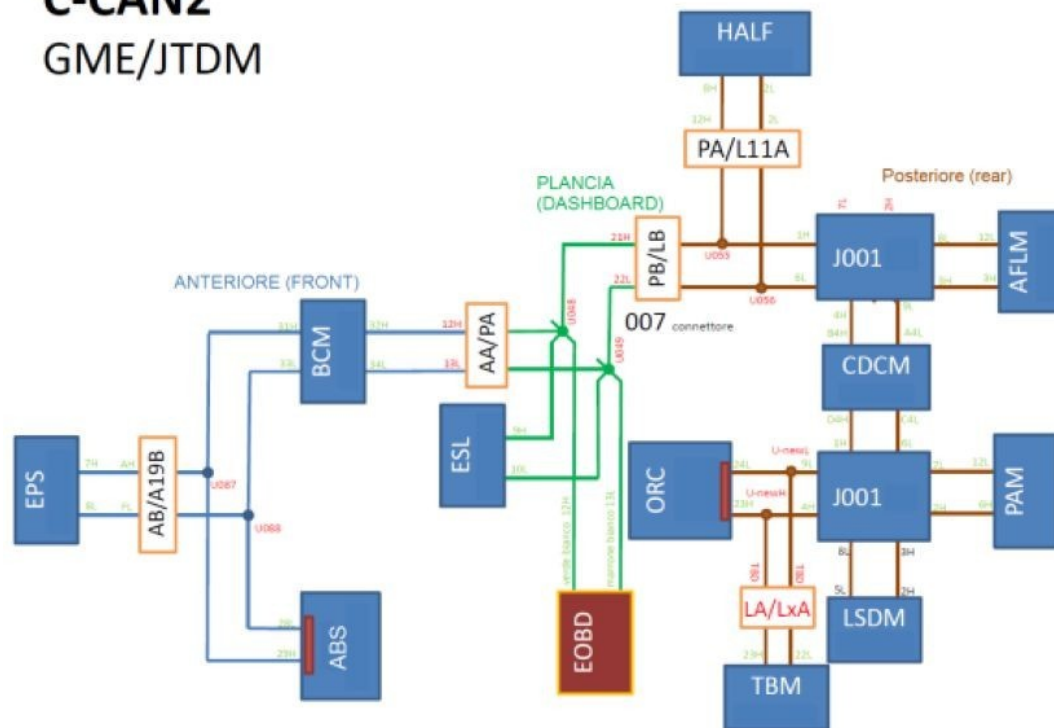
CDCM (Chassis Domain Control Module – Modulo di controllo dinamico della vettura.)

TVM (Torque Vectoring Module - Modulo di controllo del differenziale Torque Vectoring)

J001 (connettori di derivazione)

EOBD (Presa Di Diagnosi)

C-CAN2 GME/JTDM



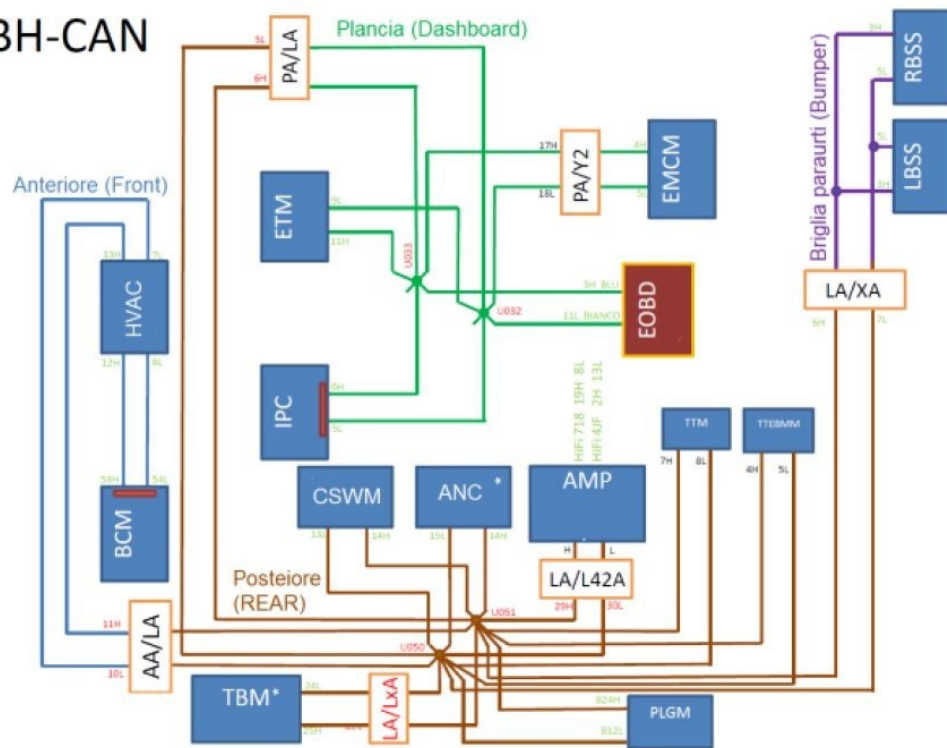


CAN-BH

The CAN-BH network is a place for data exchange for the electronic modules (ECU) that manage interior comfort:

- IPC (Instrument Panel Cluster – Quadro strumenti)
- CSWM (Comfort Seat Wheel Module – Modulo memoria e riscaldamento Sedili Volante)
- LBSS (Left Blind Spot Sensor – Sensore Blind Spot sinistro)
- RBSS (Right Blind Spot Sensor – Sensore Blind Spot destro)
- AMP (Amplifier – Amplificatore impianto HI-Fi)
- TTM (Trailer Tow Module – Modulo Gancio Traino)
- TTEBM (Trailer Tow Electric Bar Module – Modulo Gancio Traino Elettrico)
- HVAC (Heating Ventilation and Air Conditioning – Modulo clima)
- ETM (Entertainment Telematic Module – Modulo Infotelematico VP2, VP4)
- BCM (BCM – Body Control Module)
- TBM* (Telematic Box Module – Modulo Telematico)
- EMCM (Entertainment Multimedia Control Module – Selettore rotativo di gestione comandi infotelematici)
- ANC (Modulo Incremento Suono Motore)
- PLGM (Power Lift Gate Module – Modulo Portellone Baule Elettrico)
- EOBD (Presa Di Diagnosi)

BH-CAN

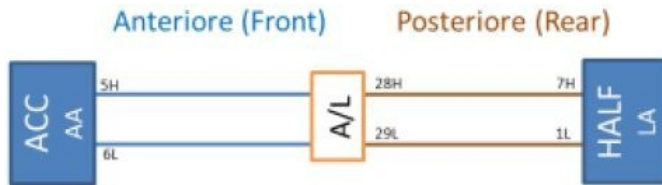




Private CAN

The ACC module is connected with a dedicated CAN-C line with the Half module. The reason for the presence of a dedicated data transmission line between the two modules is due to the continuous exchange of information between them during the operation of the adaptive cruise control and the FCW function.

PRIVATE CAN





BCM is LIN master for:

- **Lin1**
 - CHDM* (Child Detection module – Modulo per la rilevazione di bimbo a bordo)
 - IBS (Intelligent Battery Sensor – Sensore IBS polo negativo batteria)
 - UAM (Ultrasonic Antitilt Module – Modulo ultrasuoni anti-sollevamento)
 - ASU (Alarm System Unit – Allarme)
 - RLS (Rain Light Sensor – Sensore pioggia)
- **Lin2**
 - WWSM (Windshield Wiper Smart Motor – Motore elettrico del dispositivo tergicristalli)
 - SWS (Steering Wheel Switch bank – Comandi al volante)
 - RVCN (Rear View Camera Module – Videocamera posteriore)
- **Lin3**
 - WSMP (Window Smart Motor Passenger – Motorino elettrico alzacristallo passeggero)
 - WSMD (Window Smart Motor Driver – Motorino elettrico alzacristallo guidatore)
 - WSMRL (Window Smart Motor Rear Left – Motorino elettrico alzacristallo posteriore sinistro.)
 - WSMRR (Window Smart Motor Rear Right – Motorino elettrico alzacristallo posteriore destro)
 - PMCM (Passenger Mirror Control Module – Modulo di controllo specchio retrovisore lato passeggero)
 - DMCM (Driver Mirror Control Module – Modulo di controllo specchio retrovisore lato guida)
- **Lin4**
 - CSWC (Cruise control Steering Wheel Commands – Comandi cruise control al volante)

ECM is LIN master for:

- IAM (Intelligent alternator module – Alternatore intelligente)
- AUW (Auxiliary Water Pump – Pompa acqua ausiliaria)

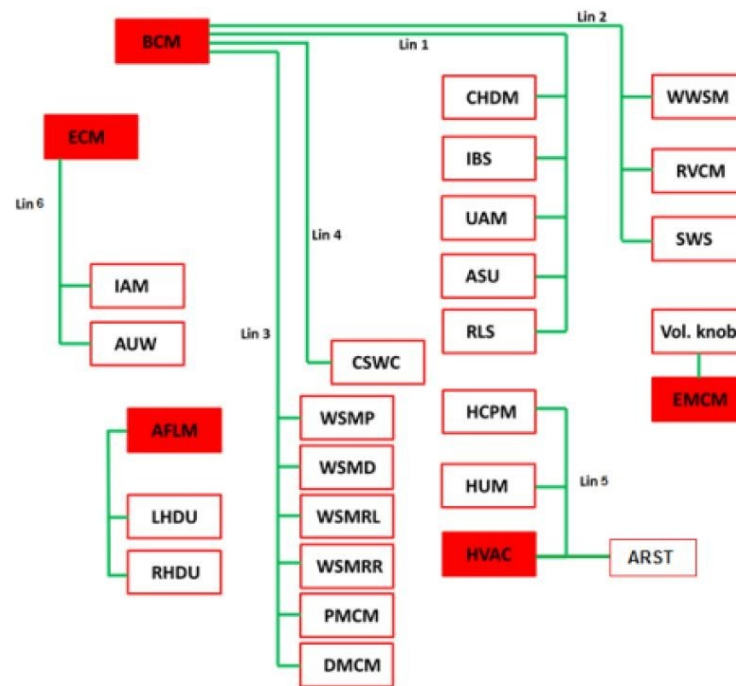
AFLS is LIN master for:

- LHDU (Left Headlamp Discharge Module – Modulo faro allo Xenon sinistro)
- RHDU (Right Headlamp Discharge Module – Modulo faro allo Xenon destro)

HVAC is LIN master for:

- HUM (Humidity Sensor – sensore di umidità)
- HCPM (HVAC Control Panel Module – Plancetta comandi clima)
- ARST (Average Radiant Temperature Sensor – Sensore Temperatura Abitacolo)

EMCM is LIN master for:





OBD-II Port

Multiple DLC Diagnosis Connector – OBD-II or EOBD

The diagnostic multiple connector pins dedicated to the three networks are:

Pin 3 - CAN-BH High

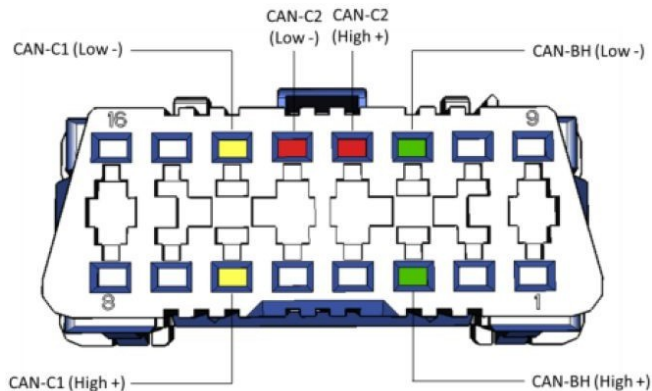
Pin 11 - CAN - BH Low

Pin 12 - CAN-C2 High

Pin 13 - CAN-C2 Low

Pin 14 - CAN-C1 Low

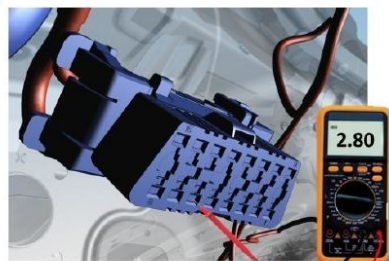
Pin 6 - CAN-C1 High



CAN Test

Voltage levels Digital networks. The voltage levels of the three networks can be read directly on the diagnostic connector using the multimeter in Ohm function.

CAN-C1



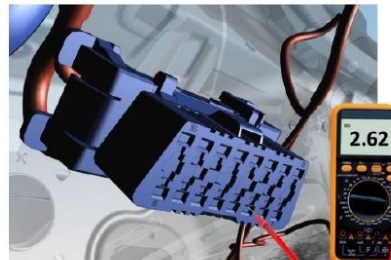
Pin 6 – Can-C1 (High Voltage)

CAN-C2

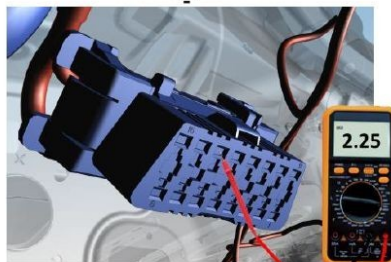


Pin 12 – Can-C2 (High Voltage)

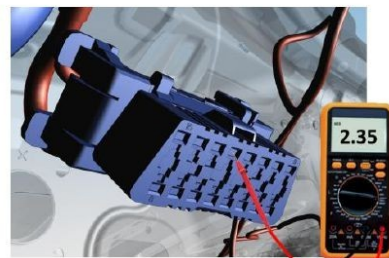
CAN-BH



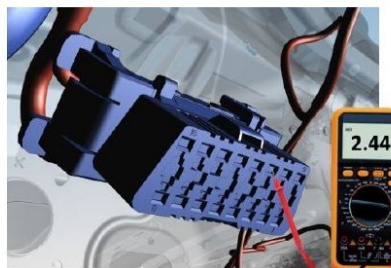
Pin 3 – Can-BH (High Voltage)



Pin 14 – Can-C1 (Low Voltage)



Pin 13 – Can-C2 (Low Voltage)



Pin 11 – Can-BH (Low Voltage)

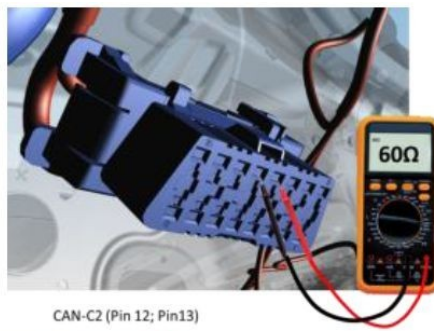
CAN Test

Electrical continuity of networks.

The three digital networks CAN-C1, CAN-C2 and CAN-BH have 120 Ω terminating resistors. When the network that interconnects the two terminating resistors is electrically continuous, the value that the operator reads on the display is about 60 Ω for all three networks.



CAN-C1 (Pin 14; Pin6)



CAN-C2 (Pin 12; Pin13)



CAN-BH (Pin 11; Pin3)

CONTACTS



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80146 Napoli

www.kineton.it





Network Topology Powernet

○ Example: DT

○ Key Features:

- **Star** topology, BCM central hub
- 11bit CAN IDs
- LIN 2.x protocol
- **NO** direct PROXI in ECM (only Vehicle Configuration via CAN)
- IPC stays on both B-CAN+C-CAN
- Cruise commands LIN connected to BCM
- Signal key architecture
- Advanced frame security mechanism (CRC/MC coverage)
- Torque CAN interface at flywheel torque level (in Nm)
- Multiple C-CANs + **Multiple** LINs
- Secure Gateway (SGW) for Cyber Security Level3
- Dedicated C-CAN for e-PT (**HEV**)

